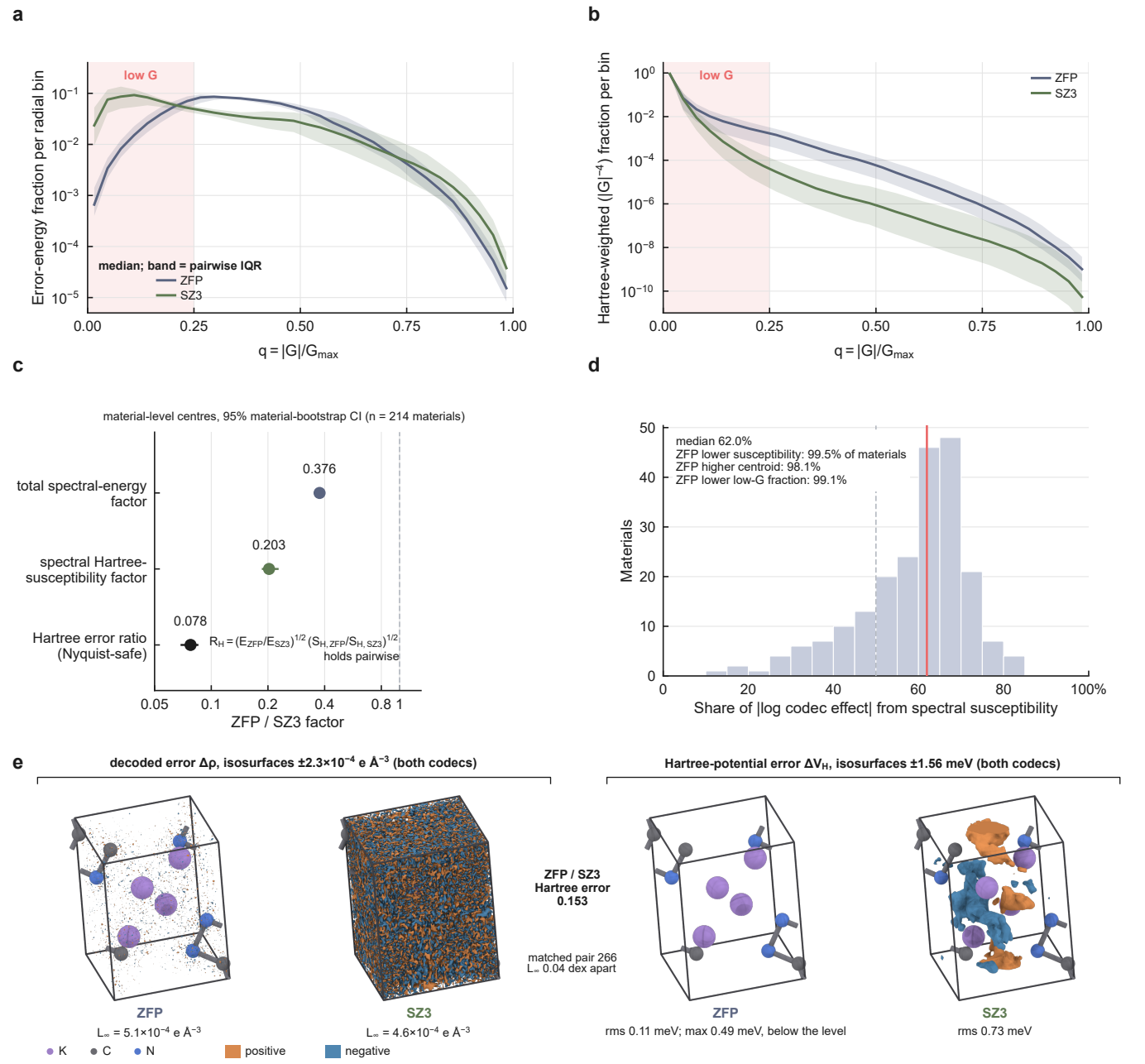

a
low G
median; band = pairwise IQR
ZFP
SZ3
Error-energy fraction per radial bin
q = |G|/G_max
b
low G
ZFP
SZ3
Hartree-weighted (|G|^-4) fraction per bin
q = |G|/G_max
c
material-level centres, 95% material-bootstrap CI (n = 214 materials)
total spectral-energy factor
0.376
spectral Hartree-susceptibility factor
0.203
Hartree error ratio (Nyquist-safe)
0.078
R_H = (E_ZFP/E_SZ3)^1/2 (S_H,ZFP/S_H,SZ3)^1/2
holds pairwise
ZFP / SZ3 factor
d
median 62.0%
ZFP lower susceptibility: 99.5% of materials
ZFP higher centroid: 98.1%
ZFP lower low-G fraction: 99.1%
Materials
Share of |log codec effect| from spectral susceptibility
e
decoded error Δρ, isosurfaces ±2.3×10⁻⁴ e Å⁻³ (both codecs)
Hartree-potential error ΔV_H, isosurfaces ±1.56 meV (both codecs)
ZFP / SZ3 Hartree error 0.153
matched pair 266
L∞ 0.04 dex apart
ZFP
L∞ = 5.1×10⁻⁴ e Å⁻³
SZ3
L∞ = 4.6×10⁻⁴ e Å⁻³
ZFP
rms 0.11 meV; max 0.49 meV, below the level
SZ3
rms 0.73 meV
K
C
N
positive
negative